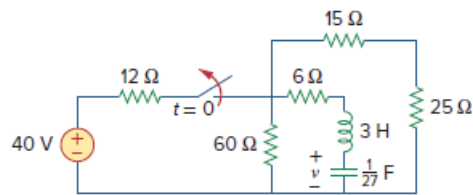


Problem

Calculate $v(t)$ for $t > 0$ in the circuit of Fig. 16.48.

Figure 16.48:



Step-by-step solution

Step 1 of 10

Refer to Figure 16.48 in the textbook.

At $t = 0^-$, the inductor acts as short circuit and capacitor acts as open circuit and the switch is closed.

Draw the circuit diagram at $t = 0^-$.

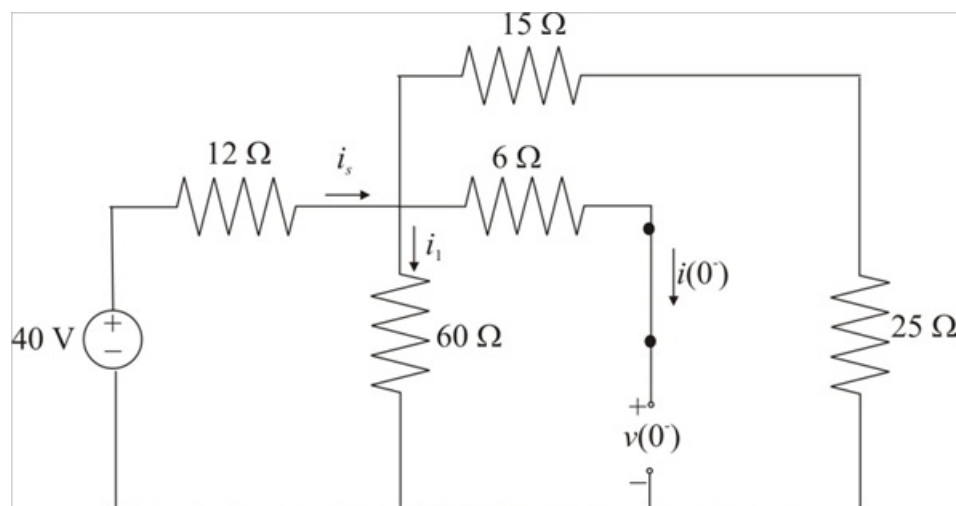


Figure 1

Step 2 of 10

The value of the inductor current is,

$$i(0^-) = 0 \text{ A}$$

Calculate the value of the current i_s .

$$\begin{aligned} i_s &= \frac{40}{12 + [60 \parallel (15 + 25)]} \\ &= \frac{40}{12 + [60 \parallel 40]} \\ &= \frac{40}{12 + \frac{(60)(40)}{60 + 40}} \\ &= \frac{40}{12 + \frac{2400}{100}} \\ i_s &= \frac{40}{12 + 24} \\ &= \frac{40}{36} \\ &= \frac{10}{9} \text{ A} \end{aligned}$$

Step 3 of 10

Calculate the value of the voltage $v(0^-)$.

$$\begin{aligned} v(0^-) &= i_s [60 \parallel (15 + 25)] \\ &= i_s [60 \parallel 40] \\ &= i_s \left(\frac{60 \times 40}{60 + 40} \right) \\ &= i_s \left(\frac{2400}{100} \right) \end{aligned}$$

Substitute $\frac{10}{9} \text{ A}$ for i_s .

$$\begin{aligned} v(0^-) &= (24) \left(\frac{10}{9} \right) \\ &= \frac{80}{3} \text{ V} \end{aligned}$$

Step 4 of 10

Draw the circuit diagram at $t > 0$ in s-domain.

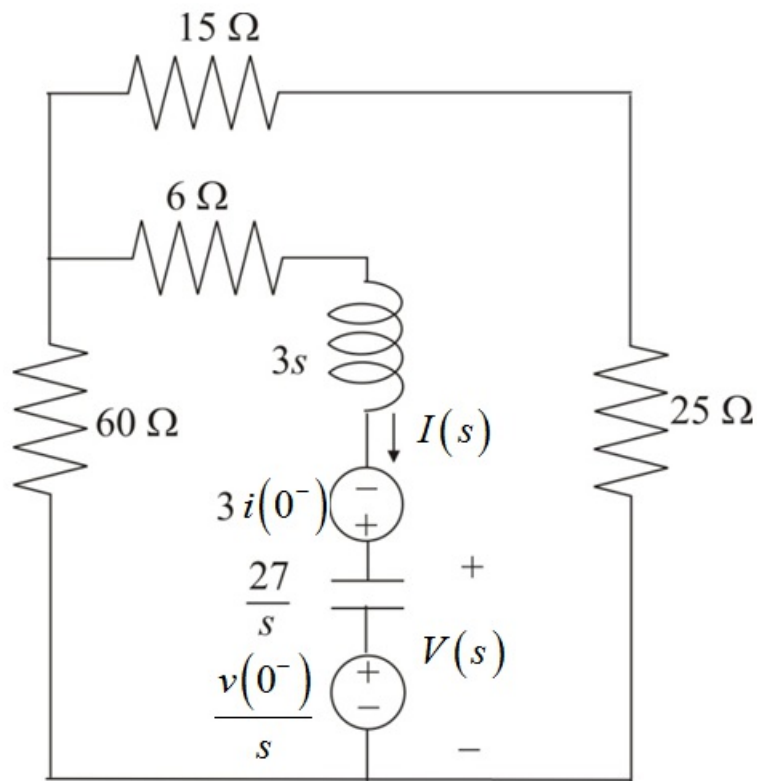


Figure 2

Step 5 of 10

Apply Kirchhoff's voltage law to the loop.

$$I(s) \left[6 + 3s + \frac{27}{s} + \frac{(60)(15+25)}{60+15+25} \right] - 3i(0^-) + \frac{v(0^-)}{s} = 0$$

$$I(s) \left(6 + 3s + \frac{27}{s} + \frac{2400}{100} \right) - 3(0) + \frac{\frac{80}{3}}{s} = 0$$

$$I(s) \left(6 + 3s + \frac{27}{s} + 24 \right) - 0 + \frac{80}{3s} = 0$$

$$I(s) \left(3s + \frac{27}{s} + 30 \right) = -\frac{80}{3s}$$

$$I(s) (3s^2 + 30s + 27) = -\frac{80}{3}$$

$$I(s) = \frac{-80}{3(3s^2 + 30s + 27)}$$

Step 6 of 10

Calculate the output voltage, $V(s)$.

$$V(s) = I(s) \left(\frac{27}{s} \right) + \frac{v(0^-)}{s}$$

Substitute $\frac{-80}{3(3s^2 + 30s + 27)}$ for $I(s)$ and $\frac{80}{3}$ for $v(0^-)$.

$$\begin{aligned} V(s) &= \left(\frac{-80}{3(3s^2 + 30s + 27)} \right) \left(\frac{27}{s} \right) + \frac{80}{s} \\ &= \left(\frac{-80}{9(s^2 + 10s + 9)} \right) \left(\frac{27}{s} \right) + \frac{80}{s} \\ &= \frac{80}{3s} - \frac{240}{s(s+9)(s+1)} \end{aligned}$$

Step 7 of 10

Apply partial fractions to the expression $\frac{240}{s(s+9)(s+1)}$.

$$\frac{240}{s(s+9)(s+1)} = \frac{A_1}{s} + \frac{A_2}{s+9} + \frac{A_3}{s+1}$$

Calculate the value of the constant, A_1 .

$$\begin{aligned} A_1 &= (s) \left[\frac{240}{s(s+9)(s+1)} \right] \Big|_{s=0} \\ &= \frac{240}{(s+9)(s+1)} \Big|_{s=0} \\ &= \frac{240}{(0+9)(0+1)} \\ &= \frac{240}{9} \\ A_1 &= \frac{80}{3} \end{aligned}$$

Step 8 of 10

Calculate the value of the constant, A_2 .

$$\begin{aligned} A_2 &= (s+9) \frac{240}{s(s+9)(s+1)} \Big|_{s=-9} \\ &= \frac{240}{(s)(s+1)} \Big|_{s=-9} \\ &= \frac{240}{(-9)(-9+1)} \\ &= \frac{240}{(-9)(-8)} \\ A_2 &= \frac{10}{3} \end{aligned}$$

Step 9 of 10

Calculate the value of the constant A_3 .

$$\begin{aligned} A_3 &= (s+1) \frac{240}{s(s+9)(s+1)} \Big|_{s=-1} \\ &= \frac{240}{(s)(s+9)} \Big|_{s=-1} \\ &= \frac{240}{(-1)(-1+9)} \\ &= \frac{240}{(-1)(8)} \end{aligned}$$

$$A_3 = -30$$

Therefore,

$$\frac{240}{s(s+9)(s+1)} = \frac{80}{3s} + \frac{10}{3(s+9)} - \frac{30}{s+1}$$

Step 10 of 10

Write the expression for output voltage in s-domain.

$$\begin{aligned} V(s) &= \frac{80}{3s} - \frac{240}{s(s+9)(s+1)} \\ &= \frac{80}{3s} - \left[\frac{80}{3s} + \frac{10}{3(s+9)} - \frac{30}{s+1} \right] \\ &= \frac{80}{3s} - \frac{80}{3s} - \frac{10}{3(s+9)} + \frac{30}{s+1} \\ &= \frac{30}{s+1} - \frac{10}{3} \left(\frac{1}{s+9} \right) \end{aligned}$$

Apply inverse Laplace transform.

$$\begin{aligned} v(t) &= 30e^{-t}u(t) - \frac{10}{3}e^{-9t}u(t) \\ &= \left(30e^{-t} - \frac{10}{3}e^{-9t} \right) u(t) \text{ V} \\ &= 0.67 \left(45e^{-t} - 5e^{-9t} \right) u(t) \text{ V} \end{aligned}$$

Therefore, the expression for the voltage $v(t)$ is $\boxed{0.67 \left(45e^{-t} - 5e^{-9t} \right) u(t) \text{ V}}$.